

Sai Srikar B

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Curriculum Vitae & Profile Summary

Undergraduate engineer in Artificial Intelligence & Data Science with an Honours specialization in Cybersecurity (Cumulative SGPA: **8.98/10**). Core research and engineering competencies in Machine Learning, Applied NLP, Local Retrieval-Augmented Generation (RAG) architectures, and AI Security (Prompt Injection Defenses & Red-Teaming). Demonstrated track record in building and deploying secure, production-grade AI platforms, optimizing on-device inference latency, and eliminating data egress risks. Seeking engineering internships and collaborative research opportunities in **Machine Learning Engineering** and **Cybersecurity**.

Education

Rajalakshmi Engineering College, Chennai <i>Bachelor of Technology in Artificial Intelligence & Data Science (Cybersecurity Honours)</i> – Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming (Java, C), Database Management Systems (SQL & PL/SQL), Operating Systems, Computer Networks, Machine Learning Foundations, Deep Learning, Natural Language Processing, Cryptography & Network Security, Linear Algebra, Probability & Statistics.	Jul 2025 – Jul 2029 SGPA: 8.98 / 10.0
Velammal New Gen Senior Secondary School, Surapet <i>Central Board of Secondary Education (CBSE) – Class 12 (Computer Science & Mathematics)</i> – Academic Distinction: Achieved a perfect 100/100 score in Computer Science with top percentile ranking in Mathematics and Physics.	Jun 2023 – May 2025 472 / 500 (94.4%)
Secondary School Examination (CBSE) – Class 10 <i>All India Secondary School Examination</i>	2023 457 / 500 (91.4%)

Technical Skills & Core Competencies

Programming Languages: Python, Java, TypeScript, JavaScript, C, SQL, PL/SQL, HTML5/CSS3
Machine Learning & NLP: Scikit-learn, PyTorch, Librosa, NumPy, Pandas, Polars, Matplotlib, Seaborn, Feature Engineering, Cross-Validation, GridSearchCV
LLM Architectures & RAG: Retrieval-Augmented Generation (RAG), LangChain, LangGraph, Ollama (Local Model Serving), Docling Layout Parsing, Prompt Engineering, Semantic Chunking
Databases & Vector Stores: PostgreSQL, SQLite, ChromaDB, Relational Schema Normalization
Web Frameworks & APIs: Next.js, React 19, FastAPI, Flask, Streamlit, Tailwind CSS, RESTful API Design, OAuth 2.0, IMAP Protocol
Cybersecurity & Defense: Metasploit, Prompt Injection Defense (Direct/Indirect), Vulnerability Assessment, Network Traffic Analysis, Kali Linux, Wireshark, SIEM Fundamentals, Port Scanning, OWASP Top 10 for LLMs
DevOps & Developer Tools: Git, GitHub Actions, Docker, Linux/Unix Shell Scripting, Markdown, LaTeX, VS Code

Technical Experience & Club Leadership

DEVS (REC) – Official Technical Club of Rajalakshmi Engineering College <i>AI / ML Associate</i> – Selected as AI/ML Associate through rigorous multi-stage coding benchmarks and algorithmic problem-solving assessments to build production software initiatives for campus-wide deployment. – Co-developed and shipped HackMentor AI and the DEVs-P2P peer-to-peer technical learning platform (learn.devsrec.com), collaborating in an agile team alongside Fullstack and DevOps leads. – Engineered security-by-design input validation guardrails preventing prompt injection exploits, conducted pre-deployment testing, and managed multi-branch Git PR review workflows.	Aug 2026 – Present Chennai, India
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Selected Engineering & Research Projects

HackMentor AI – Hackathon Evaluation Workspace & Security Guardrails – Accelerated hackathon scoring throughput by ~40% across 15+ judging panels and REC accounts by developing an AI-assisted evaluation engine with structured schema extraction. – Neutralized adversarial scoring manipulation with 0 security breaches across 30+ simulated direct and indirect injection vectors via multi-stage input sanitization, file-size boundary checks, and system prompt anchoring. – Conducted pre-deployment stress-testing during live campus event preparations, eliminating race-condition errors and ensuring reliable database synchronization.	<i>Production Platform</i>
DocMind – Decoupled Document AI & Local RAG Platform – Eliminated third-party data egress risks (100% local privacy) with sub-second retrieval latencies across complex multi-page PDF documents using Docling layout parsing and ChromaDB vector indexing. – Cut cloud API inferencing costs by ~60% by engineering an adaptive dual-engine router that dispatches routine tasks to local Ollama models and escalates multi-hop reasoning to NVIDIA NIM / OpenAI endpoints.	<i>GitHub</i>

- Implemented persistent SQLite session management and metadata filtering, enabling auditable search history across complex corporate documents and technical specifications.

MailShield-AI – Forensic Email Threat Intelligence Platform (SIH 2026)

[GitHub](#)

- Architected an 8-stage digital forensics pipeline for SIH 2026 under AICTE Cyber Security guidelines, delivering deterministic threat scoring (0–100), RFC 5322 header inspection, and sender IP geolocation for SOC analysts.
- Engineered an AI investigation copilot and graph correlation engine linking threat indicators (IPs, domains, hashes), sustaining 100% local execution without cloud telemetry.
- Built with FastAPI backend forensics, React dashboard, OAuth 2.0 Gmail ingestion, and automated structured forensic report exports.

Risonanza – Real-Time Acoustic Emotion & Stress Classifier

[Live Demo](#)

- Built and deployed an end-to-end acoustic emotion recognition service within 24 hours, achieving **~85% classification accuracy** with sub-300ms audio inference across 6 emotional states.
- Streamlined speech signal processing with Librosa MFCC and Chroma feature extraction, integrated with an optimized Scikit-learn classifier into an interactive Streamlit UI for live waveform analysis.

First-Principles ML Portfolio – Algorithm Implementations & Pipelines

[GitHub](#)

- Boosted classification accuracy from baseline to **~90.5%** across 8 iterative versions on the UCI Sonar dataset by engineering SVM classifiers from scratch and eliminating pipeline data leakage.
- Constructed reproducible Scikit-learn pipelines with StandardScaler isolation, StratifiedKFold cross-validation, and GridSearchCV hyperparameter tuning on benchmark medical datasets.

Data Structures & Algorithms Engineering Repository

[GitHub](#)

- Maintained solutions and time/space complexity analyses for core algorithms and data structures across Blind 75 / NeetCode 150 benchmarks in Python, focusing on optimal algorithmic patterns.

Certifications & Verified Credentials

Google Cybersecurity Professional Certificate – Google / Coursera (Comprehensive hands-on training in network security, Python security automation, Linux command line, SIEM tools, and incident response).

Ethical Hacking with Kali Linux – IBM Professional Credential (Penetration testing fundamentals, network reconnaissance, Wireshark packet capture analysis, vulnerability scanning).

Artificial Intelligence Foundations – LinkedIn Learning (Core machine learning principles, neural network architectures, and algorithmic decision models).

Honors & Extracurricular Highlights

Academic Centum Award: Scored a perfect **100/100 in Computer Science** in CBSE Class 12 Board Examinations.

Top Undergrad Academic Standing: Consistently maintained **SGPA 8.98/10.0** in B.Tech AI & Data Science (Cybersecurity Honours).

Hackathon Rapid Prototyping: Built and deployed full-stack speech processing applications (Risonanza) within strict 24-hour sprint limits.

Technical Club Competitive Selection: Selected as AI/ML Associate at DEVS (REC) through multi-round competitive technical evaluations.